

HW :- 22/07/26.

Q) Java generates hash code for every object. Find out what is the hash code in java and the purpose of generating Hash code.

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In Java, a hash code is an integer value associated with an object. It is returned by the `hashCode()` method. It is used to identify and locate objects quickly, especially in hash based collection like "HashMap" and "HashSet".

Why does Java generate hash code?

Imagine you have a library of 10,000 books.

If someone asks for one book - then without hash code - you search every shelf one by one.

with hash code - you know exactly which shelf to check.

Similarly in Java, it helps store and retrieve objects faster.

Hashcode acts as a roll number for the object which improves the program performance.

Memory Address - Actual location i.e. RAM (hidden from programmers)

hash code - Integer value used for searching & storing of objects, it is not the memory address.

Example :-

```
public class Main {  
    public static void main(String[] args) {  
        String s = "Hello";  
        System.out.println("s.hashCode()");  
    }  
}
```

Output :- 69609650